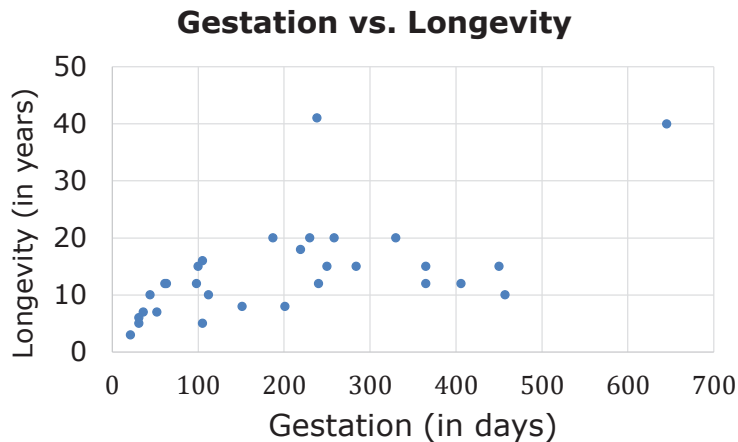




### 12.5.6 Wrap-Up: Ticket Out the Door

1. The following scatter plot summarizes the average gestation period, in days, and average longevity, in years, for a sample of 30 animals as reported in *The World Almanac and Book of Facts* (McGeveran, 2006). A gestation period is the amount of time for which an animal is pregnant. Longevity refers to how long an animal lives.



- a. What does correlation mean in this context?
- b. Does the data provide evidence that a longer gestation period causes animals to live longer?

## Unit 12, Lesson 6: Estimating for a Population



### Benchmark

MA.912.DP.1.4 Estimate a population total, mean, or percentage using data from a sample survey; develop a margin of error through the use of simulation.



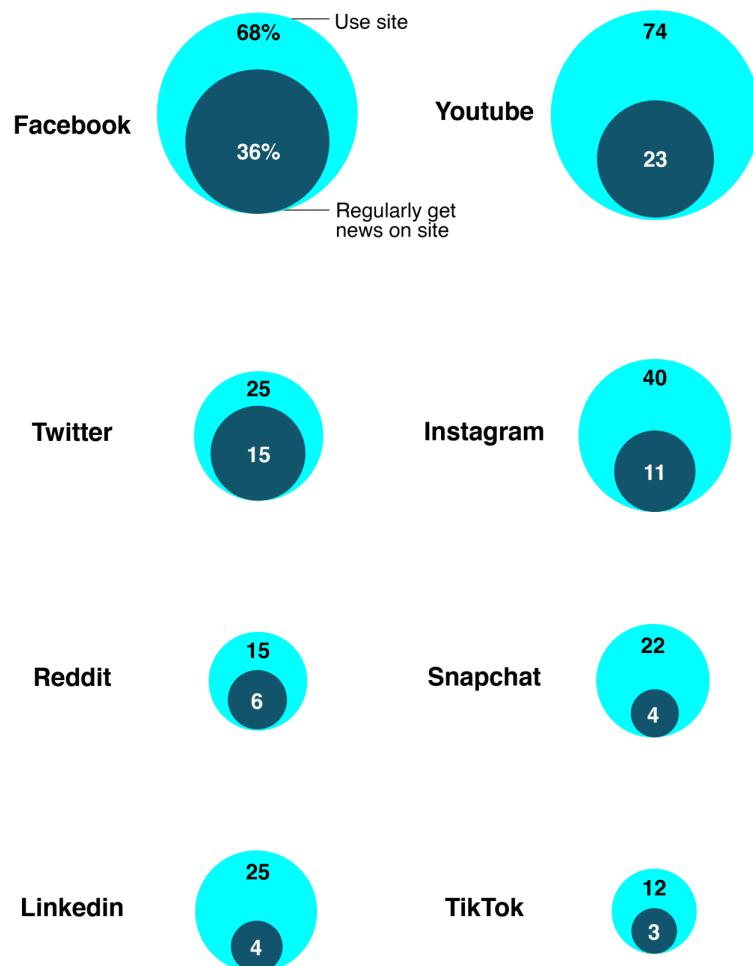
### Learning Target

- ☐ I can use margin of error to estimate a population total, mean, or percentage using data.



### 12.6.1 Warm-Up: News Source

1. The graphic below summarizes data from a 2020 Pew Research Study. It shows the percent of U.S. adults who use each site and regularly get news on each site.



What do you notice? What do you wonder?



### 12.6.2 Guided Instruction: Margin of Error

In 2020, 9,220 U.S. adults were surveyed about their news use across social media platforms.

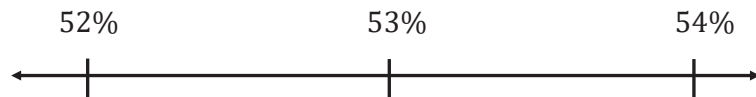
53% of those surveyed reported that they get their news from social media “often” or “sometimes.”

1. If you were able to conduct a similar survey of U.S. adults, do you think exactly 53% would report that they get their news from social media “often” or “sometimes?” Explain your reasoning.
2. Can it be concluded that exactly 53% of all U.S. adults get their news from social media “often” or “sometimes?”
3. Do you think that the majority of U.S. adults get their news from social media sometimes or often?

4. Estimate the total number of U.S. adults who got their news from social media often or sometimes in 2020 if there were 256,662,010 U.S. adults at the time of the survey.

When a population percentage or mean is estimated from sample data, it is often helpful to give a range for the estimate to consider variability in the sample.

Based on the same sample data, it can be concluded that the population percentage is between 52% and 54%.

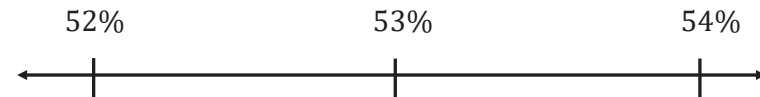


This interval of possible values for the estimate of the population percentage was created using a **margin of error**.

The **margin of error** quantifies the amount of random sampling error in the results of a survey.

In the sample data of 9,220 U.S. adults, the percentage who get their news from social media “often” or “sometimes” is 53% with a margin of error of  $\pm 1\%$ .

5. Estimate the range for the total number of U.S. adults who got their news from social media often or sometimes in 2020 if there were 256,662,010 U.S. adults at the time of the survey using the estimate of 53% with a margin of error of  $\pm 1\%$ .



$$0.52 \cdot 256662010$$

$$0.54 \cdot 256662010$$

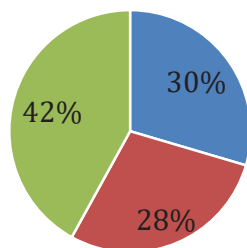
The total number of U.S. adults who get their news from social media sometimes or often is estimated to be between \_\_\_\_\_ and \_\_\_\_\_.



### 12.6.3 Guided Practice: Estimating for a Population

1. The results of 741 teens surveyed in a 2018 Pew Research Study are shown in the circle graph.

**Teen Views on the Effect of Social Media on People Their Age**



■ Mostly Positive ■ Mostly Negative ■ Neutral

- a. If there were 24.97 million teens in the U.S. population at the time of the survey, what is the estimated number who believed the effect of social media on people their age had been mostly positive?
- b. The proportion of all U.S. teens who believe the effect of social media has been neutral on people their age is 0.42 with a margin of error of 0.036. Estimate the range of all U.S. teens who believe the effect of social media has been neutral on people their age.

2. In a survey of 175 students at Satellite High School (SHS), 82 reported that they use Facebook.
  - a. The estimated proportion of all Satellite High School students who use Facebook is \_\_\_\_\_ with a margin of error of 0.074.
  - b. The estimated proportion of all Satellite High School students who use Facebook is between \_\_\_\_\_ and \_\_\_\_\_.
  - c. Do you think it's reasonable to conclude that the majority of SHS students use Facebook? Explain your reasoning.
  - d. Suppose there are 1,385 students enrolled at Satellite High School. The estimated number of all Satellite High School students who use Facebook is between \_\_\_\_\_ and \_\_\_\_\_.

3. A large company tracked the amount spent on each of their Amazon purchases over the past four years. For a sample of 135 purchases, the average amount spent was \$35.23 with a margin of error of  $\pm \$2.16$ .

If the company predicts they will make 50 purchases on Amazon in the next month, what is the range of the predicted total amount they will spend?



### 12.6.4 Your Turn!

1. A new drug was given to a random sample of 400 people with acne. The results showed that 210 of the people's acne improved and the results had a margin of error of  $\pm 0.049$ .
- a. Based on these results, can it be said that more than half the people with acne will improve if they are given the drug?

- b. If the new medication will be administered to 5,552 people, how many are expected to see improvement?

Between \_\_\_\_\_ and \_\_\_\_\_ are predicted to see an improvement in their acne.

2. During harvesting season at Florida Blue Farms, hand pickers collected 200 pounds of blueberries and found a 95% pack-out rate. The margin of error for the sample was  $\pm 3\%$ . The pack-out rate is the percentage of collected blueberries that can be packaged to be sold, based on Florida Blue Farms' quality standards.
- The estimated percentage of all blueberries that are hand-picked and can be packaged and sold is between \_\_\_\_\_ and \_\_\_\_\_.
  - If 2,800 blueberries are hand-picked, what is the most that are expected to be packaged and sold?
3. Publix Supermarkets Inc., commonly known as Publix, has 831 locations in Florida. Publix offers free cookies to children to eat while they're in the store. A survey of 100 store managers found that the average number of cookies given out on a weekday was 15.5, with a margin of error of 2.1.

Complete the statement.

The estimated total number of free cookies given out each weekday for all Florida locations is between \_\_\_\_\_ and \_\_\_\_\_.



## 12.6.5 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each.



I need help.



I got it.



I could help someone.

|                                                                                 |                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Classifying data as categorical or quantitative, and as univariate or bivariate |          |
| Creating and interpreting graphical displays for categorical data               |          |
| Creating and interpreting graphical displays for numerical data                 |          |
| Explaining the difference between correlation and causation                     |          |
| Using the margin of error to estimate population totals, means, and percentages |    |